Week 5

Messages and My Blocks

Level 1: Introduction

Break big programs into named parts.

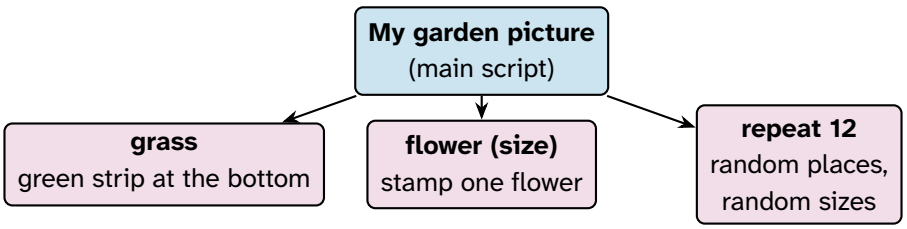
What you will learn

- How to break a big job into smaller named parts (**decomposition**).
- How to make your own block with **My Blocks**, and give it an **input**.
- How sprites send each other **messages** with **broadcast**.
- How to pick colors and text that **everyone** can see.

The big idea

Big jobs, small named parts

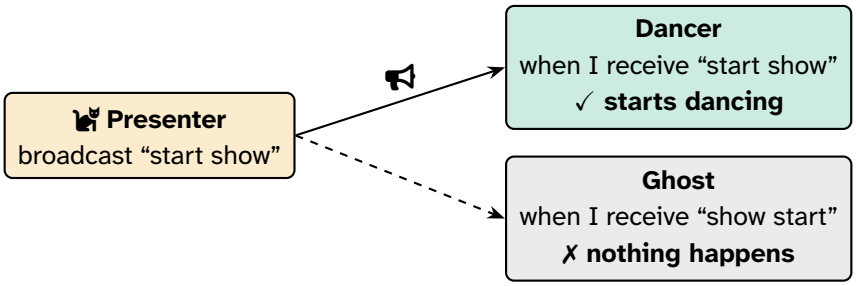
A garden picture is a big job. Break it into parts and give each part a name. Then the main plan is short and easy to read.



In Scratch, **My Blocks** → **Make a Block** lets you make a pink block with your own name. You put its steps under a **define** hat. An **input** is a blank you fill in each time you use the block, like “size”.

Messages

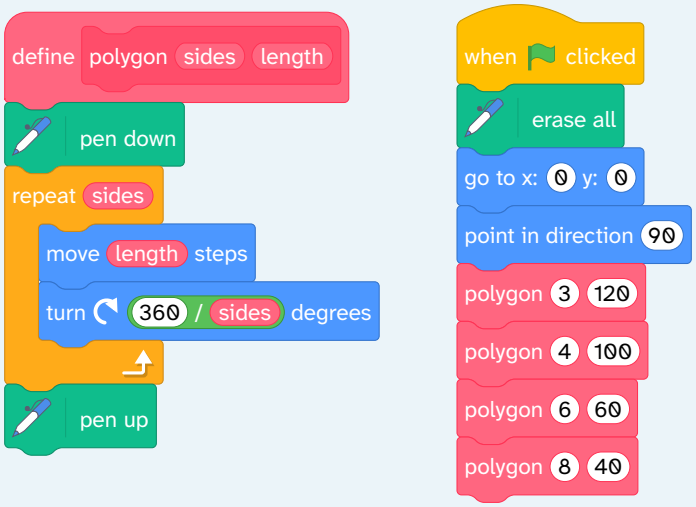
broadcast sends a message to **every** sprite, like an announcement on a loudspeaker. Only scripts that start with **when I receive that exact message** run.



Exercise 5.1: Shape Machine

Make your own block that can draw **any** shape.

1. New project. Add the **Pen** extension. Set the cat’s **Size** to 40.
2. **My Blocks** → **Make a Block**. Type `polygon`. Click **Add an input (number or text)**, type `sides`. Add another input: `length`. Click **OK**.
3. Build the define script. Drag the `sides` and `length` ovals **out of the define hat** into the slots.
4. Build the green-flag script with your new pink block. Click the flag!
5. Add a pentagon: one more block, “polygon 5 70”.

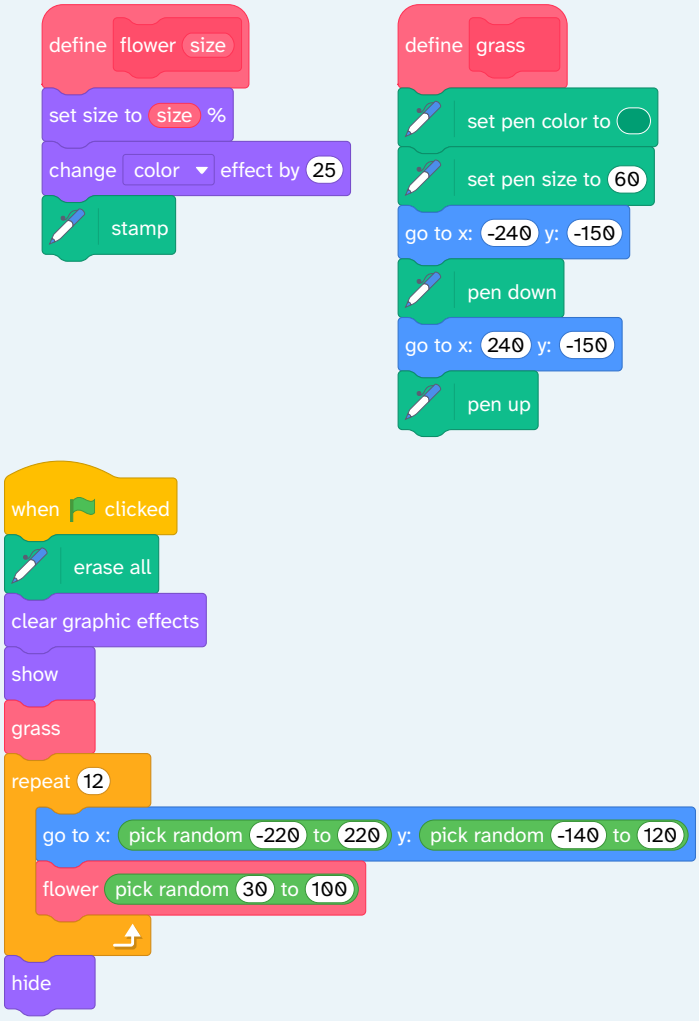


How many blocks would you need to draw these 4 shapes **without** your My Block? _____

Exercise 5.2: Stamp Garden

1. New project with the **Pen** extension. Delete the cat. Hover over **Choose a Sprite** and click the **Paint** brush. Draw a flower in the middle of the canvas (circles for the center and petals). Name the sprite `Flower`.

2. Make two My Blocks: `flower` with an input `size`, and `grass` with no input.
3. Build the three scripts. For the grass color, click the color circle and set **Color 45, Saturation 100, Brightness 62** (see the color-safe paint box).



Done when: each click on the flag grows a new garden: a strip of grass and 12 flowers of different sizes and colors in surprise places.

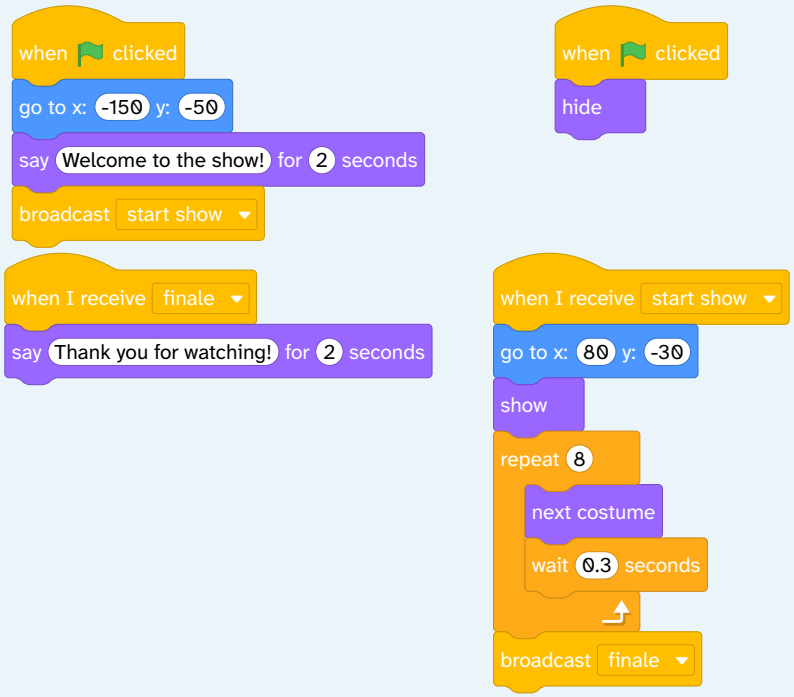
Exercise 5.3: Show Time

Two sprites put on a show by sending messages.

1. New project. Keep the cat (the **Presenter**). Add a **Dancer**: any sprite with several costumes, like Ballerina.
2. To make a message: in a “broadcast” block, click the dropdown → **New message**, type `start`

`show`. Make another called `finale`.

3. Build the scripts. Click the flag and enjoy the show!



Presenter (cat) **Dancer**

Bug hunt! Ada's dancer starts with "when I receive **show start**". Try it. What happens? Why? How do you fix it?

Digital citizenship & safety: Color-safe paint box

Some people see colors differently. These colors stay different for almost everyone. In Scratch, click a color circle and type the three numbers. And never use color as the only clue: add a shape, a pattern or a word too, and keep text big.

Color	Col.	Sat.	Bri.	Color	Col.	Sat.	Bri.
Orange	12	100	90	Blue	56	100	70
Sky blue	56	63	91	Vermillion	7	100	84
Bluish green	45	100	62	Reddish purple	91	41	80
Yellow	16	72	94	Black	0	0	0

Vocabulary box

decomposition Breaking a big problem into smaller parts.

My Block A block you make yourself, with a name you choose.

define The hat block that holds the steps of your My Block.

input A blank in a block that you fill in each time, like “size”.

broadcast Send a message to every sprite.

when I receive A hat block that runs when its exact message is sent.

accessible Easy for everyone to use, including people who see or hear differently.

Reflection

Why is “polygon 6 60” easier to read than the blocks inside it?

What did you do to make your project easier for everyone to see?

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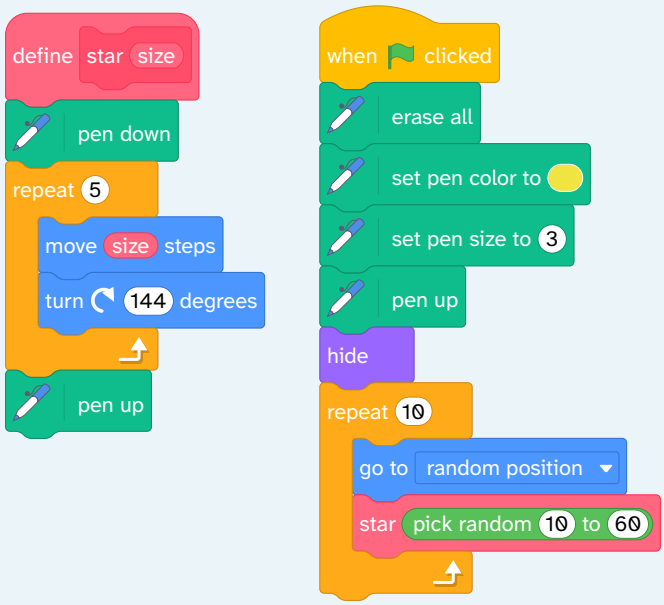
15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Starry Sky (Scratch, about 10 minutes)

Make your own “star” block and fill the night sky.

1. Open Scratch and add the **Pen** extension. Pick a dark backdrop (for example **Stars**, or paint the backdrop dark blue).
2. **My Blocks** → **Make a Block**: name it `star`, add an input `size`, click **OK**.
3. Build both scripts. Drag `size` out of the define hat into the move block. For the pen color, click the color circle and set **Color 16, Saturation 72, Brightness 94** (yellow from the color-safe paint box).
4. Click the flag. Save as `starry-sky`.



Why does the star block turn 144 degrees and not 72? (Hint: count how many times the pen goes all the way round.)

⭐⭐ Challenge (want more?): Add a moon (decomposition)

Make a second My Block called `moon` with no input. It should go to the top-left corner and draw a big circle: repeat 36, move 8, turn 10. Use it in your green-flag script before the stars. Now your main script reads like a plan: erase all, moon, stars. Write your main script as a plan, one line per block:

🎨 Creative (make it yours): Good night message

Add a **Button** sprite (any sprite) that says “Click me for night time”. When it is clicked, it broadcasts a new message `night falls`. Give the cat a “when I receive night falls” script that says “Good night!”. Check that both blocks use exactly the same message.

✅ Self-check: tick each box when it is true

- ☐ My `star` block has an input called `size`, and I used it inside.
- ☐ Every click on the flag draws 10 stars of different sizes in different places.
- ☐ My stars can be seen clearly on the dark sky (I used a color from the paint box).
- ☐ (Creative) My broadcast and my “when I receive” use the same message.